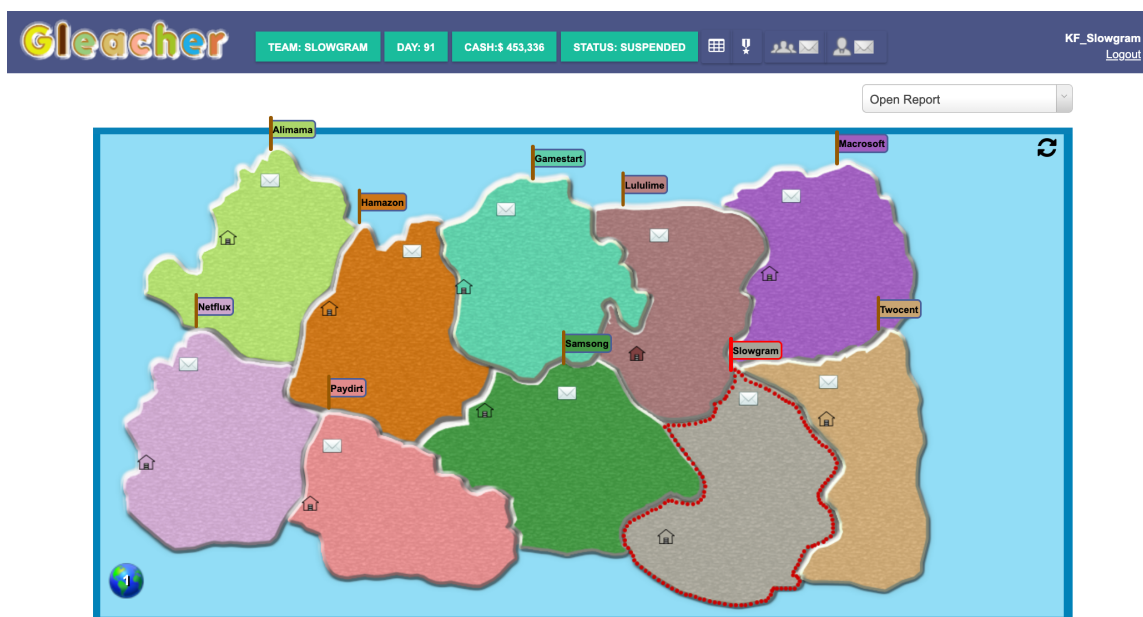


Monopoly Game Case Brief

Overview

Welcome to the Monopoly Game. Your team will be managing a firm that manufactures and sells two wearable medical monitors as a monopolist in its own region. The first product, named “Hormone”, tracks the concentration of several hormones. It is a generic product that any firm can make. The second product is a proprietary specialty good.

After logging in to the game, you will see:



The top row shows your team name, the game day, the current cash balance, the game status, your username, and the logout button. The game status may be suspended, running, or done.

Your home region is outlined in red dots on the map and contains a distribution center (DC) icon that opens a panel to manage inventory and the pricing of your hormone and specialty products.

There are four controls above the map:

- The “History” button opens a table showing all the actions taken by your team.
- The “Scoreboard” button shows all the teams ranked by cash position.
- The “Chat” buttons allow open chat windows for you to communicate with all teams or admin.
- The “Open Report” dropdown opens a list of useful reports (see the *Reports* section below).

Background

The firm manufactures and sells two wearable medical monitors. It was founded 91 days ago by an engineer who designed the products. He contributed his life savings of \$3,000,000 to the firm and used the funds to purchase a factory for production and a distribution center to store and ship the products. He was struggling with business management, when you and your team came along and paid him \$1,500,000 for 50% of the firm. Your team will now take over managing the business.

Products are manufactured in batches of 100. The materials cost for each product is \$100. Production of a batch begins by incurring a cost of $\$100 * 100 \text{ units} = \$10,000$. The batch takes 2.5 days to manufacture, and the factory can only make one batch of a single product at a time. Once the batch is completed, another batch can begin production while the finished batch goes into “transit” for 24 hours as it is sent from your factory to your DC. Shipping from your factory to your DC is free. Once the batch arrives at your DC, it goes into on-hand inventory. All sales are made from on-hand inventory at your DC. The factory and DC operate 24 hours a day.

There is a shipping agreement for each product that is manufactured at the factory and sent to your DC. You control production by setting a reorder point in the shipping agreement for each product. The DC asks the factory to make a new batch when the sum of in-process, in-transit, and on-hand inventory is equal to or less than your reorder point. You can turn off production by setting the reorder point to -1. If production of both products is requested simultaneously, then the facility alternates producing a batch of one product then a batch of the other product.

The market size for the hormone product is 300,000. The market size for the specialty product is 140,000. Retail customers for each product arrive at the DC daily at a constant rate of $0.0001 * \text{the market size}$. The willingness to pay (WTP) of individual customers is uniformly distributed along a range of \$0 to maximum WTP. You set a separate retail price for each product. If a customer for a particular product arrives and that product’s price is higher than what the customer is willing to pay, the customer is lost forever. If a customer for a particular product arrives and there is no on-hand inventory to sell, the customer is also lost forever.

At the end of the game the monitor technology will be replaced by a new generation of technology. As result the company shuts down and all inventory becomes obsolete. Your objective is to maximize the firm’s cash balance at the end of the game.

Distribution Center Controls

Click the DC icon  in your region to open a panel to manage the inventory for all of your products. You will see the following icons:

-  opens a panel to show more details
-  refreshes the panel
-  next to a column label reveals additional columns of data
-  lets you to change a parameter
-  opens a daily plot of some metric
-  downloads data from a report

Change the retail price for any of your products by clicking  next to retail price in the Inventory table of the DC.

Change the reorder point for any of your products by clicking  next to the shipping agreement in the Incoming Shipping Agreements table.

Click  in the batches column to open a table of all the batches created and shipped under that agreement. You can only make changes while the game is running.

Communication

You may communicate through the chat buttons on the top of the map page or by clicking on the envelopes in other regions.



Use this to send a group chat.



Use this to send a chat to the admin team.

Click on the envelopes in each region to send a message to that region's team.

If the envelope in your region is yellow, it means that someone has sent you a message.

Other Considerations

If your firm runs out of cash it will receive emergency loans to continue operations. Emergency loans are automatically repaid as cash becomes available. Interest on emergency debt is charged at an annual percentage rate (APR) of 40%.

Cash on your balance sheet earns interest at a 3% APR.

Cost of capital is 10% APR.

Expenses for raw materials are payable 15 days after they are incurred.

Profits are taxed at a rate of 35%, payable at the end of every quarter.

Reports

The **Financial Statements** report contains the firm's consolidated income statement, balance sheet, and cash flow statement. Use the dropdowns at the top of the report to select the team, statement, and time period: days, weeks, quarters, and years. Only quarterly and annual statements are available for other teams. Click the \$ symbols in the balance sheet or income statement to switch to a % view and vice versa. Clicking the \$/% at the top of the column switches the entire statement. Download a statement by clicking  at the top right of the page.

The **Working Capital Analysis** report contains the firm's current assets, current liabilities, total working capital and current ratio, non-cash working capital and current ratio, and quick ratio. Use the dropdowns at the top of the report to select the team and time period: days, weeks, quarters, and years. Only quarterly and annual information is available for other teams. Click on an item label to see a chart of the values for that item. Download the report by clicking  at the top right of the page.

The **Market Grid** report contains matrices of various metrics for each product market and geographic region in the game. The metrics are cumulative unit sales, cumulative revenues, average retail price, fraction of the market served, and concentration of sales by team. Select a metric using the dropdown at the top of the report. Download the data by clicking  at the top right of the page.

The **Price Response** report provides data on average retail units sold per day at each price for a given product in a DC, as well as the ability to plot various relationships in a graph above the data table. Switch to a different product by clicking  next to the product name at the top of the page. Download the data by clicking  at the top right of the page.

The **Inventory Status** report shows a snapshot of your supply chain at a given point in time: batches being produced at the factory, batches in transit to the DC, on-hand inventory levels at the DC, stocked out items, and inactive inventory. The data includes product names, locations, quantities, days remaining until new batches are produced and delivered, and estimated days remaining until on-hand inventory runs out.